ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 4: Quadratic Equations (Dvighat Samikaran)

Key Concepts & Important Formulas:

* Standard Form: ax^2 + bx + c = 0, where a != 0.

* Methods of Solving: 1. Factorisation (splitting middle term), 2. Quadratic Formula.

* Quadratic Formula (Shridharacharya Rule): x = [-b +- sqrt(b^2 - 4ac)] / (2a).

* Nature of Roots via Discriminant D = b^2 - 4ac:
- If D > 0: Two distinct real roots.
- If D = 0: Two equal real roots (x = -b/2a).
- If D < 0: No real roots.

Solved Exam Examples & Practice:

Example: Find discriminant of 2x^2 - 4x + 3 = 0 and nature of roots.
D = (-4)^2 - 4(2)(3) = 16 - 24 = -8 < 0.
Hence, equation has no real roots.